

# Employee Future Performance

## Random Forest Model Evaluation Report

Model type: RandomForestClassifier

Test set size: 200 records

**0.985**

Accuracy

**0.986**

Precision

**0.985**

Recall

**0.985**

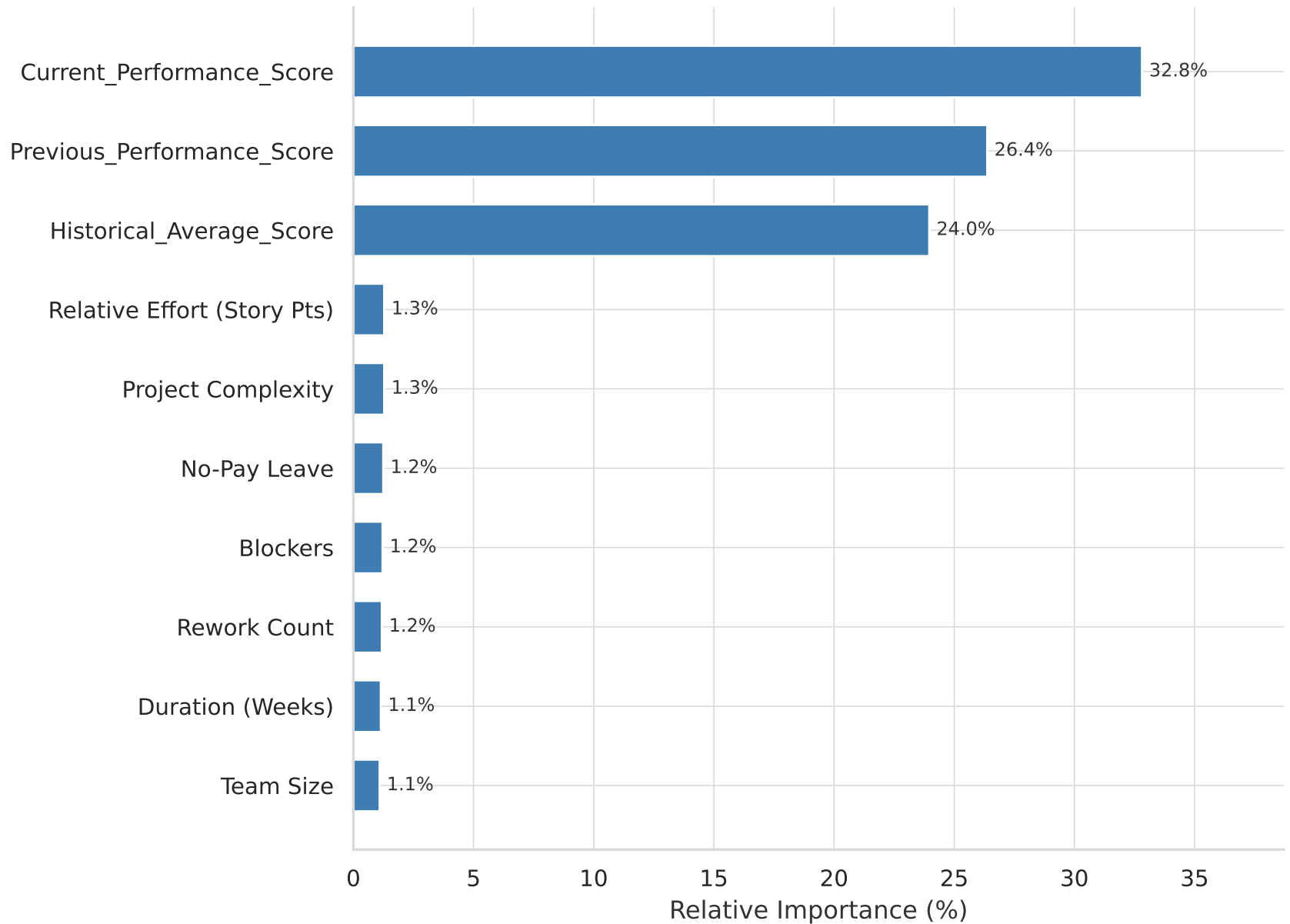
F1 Score

**0.999**

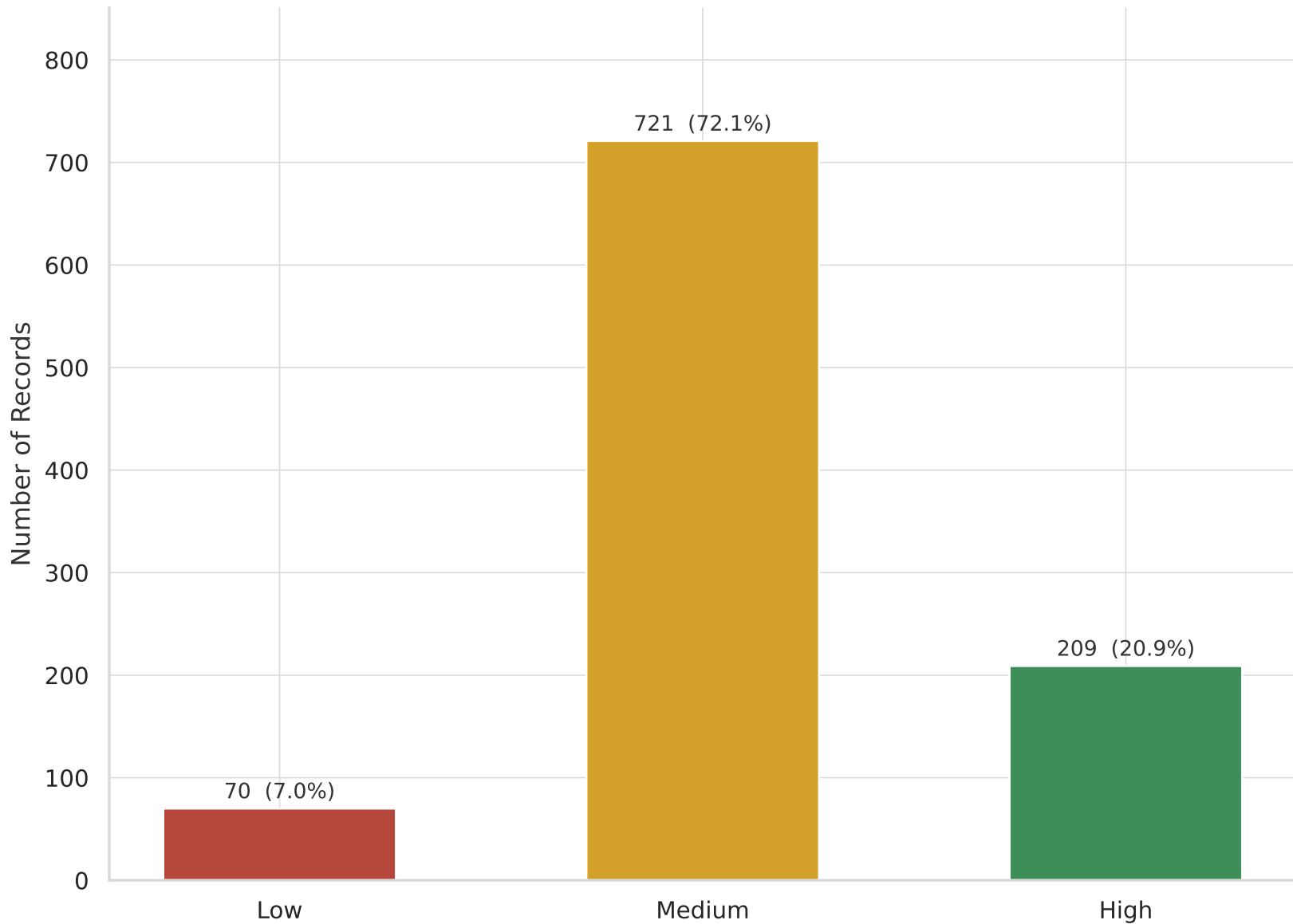
ROC-AUC

This report summarizes classifier performance for predicting an employee's performance band (Low / Medium / High) in the next quarter, based on current-quarter KPIs and history.

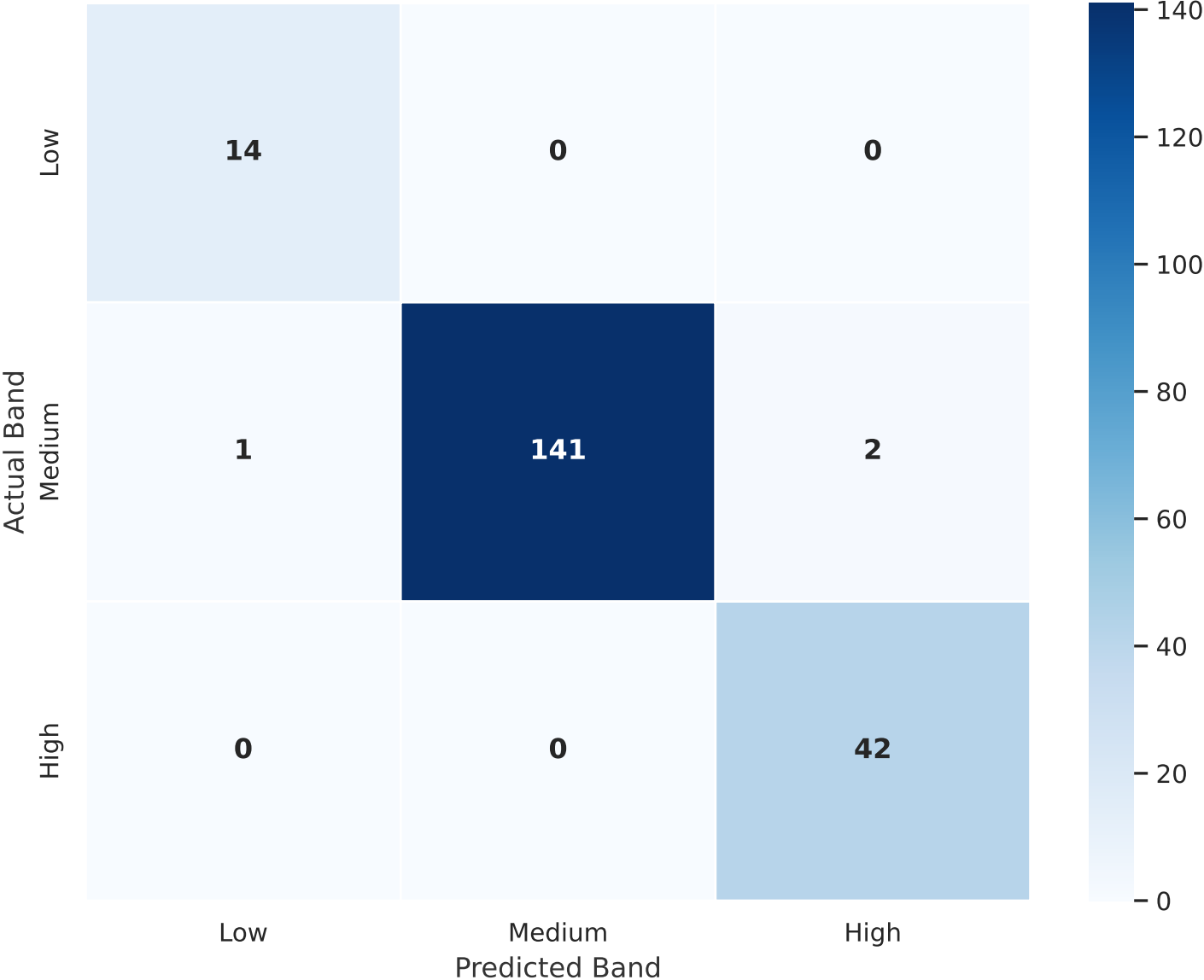
## Top 10 Most Influential Features



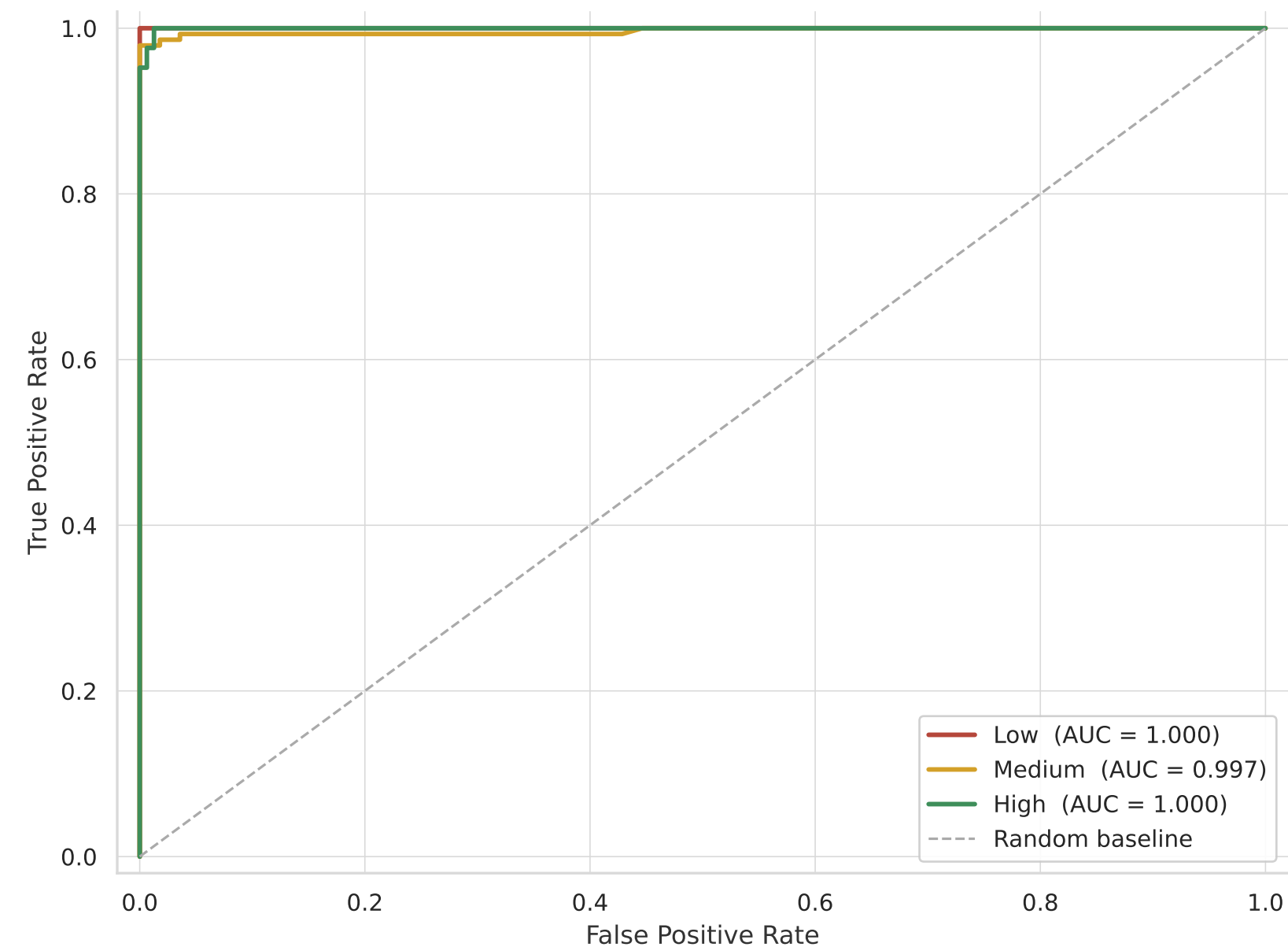
## Future Performance Band — Class Distribution



Confusion Matrix — Predicted vs. Actual



ROC Curves — One-vs-Rest by Performance Band



Per-Class Performance Summary

| Band         | Precision | Recall | F1-Score | Support |
|--------------|-----------|--------|----------|---------|
| Low          | 0.933     | 1.000  | 0.966    | 14      |
| Medium       | 1.000     | 0.979  | 0.989    | 144     |
| High         | 0.955     | 1.000  | 0.977    | 42      |
| Macro Avg    | 0.963     | 0.993  | 0.977    | 200     |
| Weighted Avg | 0.986     | 0.985  | 0.985    | 200     |

Overall Accuracy: 0.985 | Weighted F1: 0.985 | ROC-AUC: 0.999